Name _____ Date ____ / ____ / _____ Score _____

SCIENCE · GRADE 8 · WORKSHEET

Newton's Three Laws of Motion

Read the story, then identify the laws and use the formula. Answer key on the last page.

A Brief History

Every question tells you which paragraph to look in, and the vocabulary words are in **bold** in the story.

1 The skateboard rolled away without him, and that was the first clue. Marcus had stepped off it to tie his shoe, and the board just kept going. Straight across the flat concrete, past the bench, slowing a little but not stopping, until it clipped the lip of the ramp and finally gave up. Nobody had pushed it. It simply carried on doing what it had already been doing.

2 That is the first thing to understand about the world: it is lazy. Objects continue exactly as they are until something interferes. A board that is rolling keeps rolling. A board sitting still sits still all afternoon. The word for that stubbornness is **inertia**, and the only thing that beats it is a push or a pull — a **force** — strong enough to win. If the forces on something are balanced they cancel out and nothing changes. Only an **unbalanced** force changes anything. That is **Newton**'s first law.

3 Marcus picked up his board and set it next to his friend Dev's. Dev's board was the heavy kind, thick trucks, big wheels, built like a door. Marcus pushed his own board with a gentle shove and it shot away. He pushed Dev's with exactly the same shove and it lumbered off like it resented being asked. Same push, very different results. The difference was **mass** — how much stuff each board was made of. What changed was **acceleration**, which does not only mean speeding up; it means any change to how fast something is going or which way it is going. Newton wrote that down as F = m × a, and it is the second law: push harder and you get more acceleration, add mass and you get less.

4 Then Marcus did the thing he had done ten thousand times without thinking about it. He put one foot on the board, and with the other he pushed backward against the ground — and he went forward. Read that again, because it is stranger than it sounds. He pushed backward. He traveled forward. His foot shoved the whole planet one way and the planet shoved him the other. Forces are never lonely; they arrive in pairs, equal in strength and opposite in direction. That is the third law, and it sounds like a riddle until you notice you have been using it your whole life.

5 It is also the law that got us off the ground. A wing works by shoving air downward, and the air shoves the wing up. A jet engine throws air out the back and the plane is thrown forward. A rocket in the emptiness of space has nothing at all to push against, and does not need anything — it pushes its own exhaust one way and rides the other way on the reaction.

6 About that formula, because the questions below use it. F stands for force, measured in newtons and written N. The m is mass in kilograms, kg. The a is acceleration in meters per second squared, m/s². If you know two of the three you can always find the third, and there are only three versions to know: F = m × a, or a = F ÷ m, or m = F ÷ a. The rule underneath all of them is short — if you are looking for F you multiply, and if you are looking for anything else you divide F by the other one. Worked example: a cart with a mass of 5 kg accelerates at 2 m/s², so F = 5 × 2 = 10 N. Always write the unit. An answer of 10 on its own is not finished.

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Vocabulary — write what each word means

force _____ **inertia** _____

mass _____ **acceleration** _____

unbalanced _____ **newton** _____

Eight Questions — answer in your own words

Before you answer. Every answer is in the story. Go to the paragraph each question names, find the sentence that answers it, and write its question number in the margin beside it. The last three use the formula, and paragraph 6 shows you how with a worked example. Write the unit on every answer.

1. The skateboard kept rolling after Marcus stepped off it. Which law explains that, and why? **(Paragraphs 1 and 2)**

2. Marcus gave both boards the same push, but Dev's moved off more slowly. Explain why, using the word *mass*. **(Paragraph 3)**

3. Marcus pushed backward against the ground and traveled forward. Explain how that works. **(Paragraph 4)**

4. A rocket in empty space has nothing to push against. Explain how it still moves forward. **(Paragraph 5)**

5. What does *inertia* mean? **(Paragraph 2)**

6. A cart has a mass of 4 kg and accelerates at 3 m/s². What force was applied? Show your work. **(Paragraph 6)**

7. A force of 50 N acts on a 10 kg box. What is its acceleration? Show your work. **(Paragraph 6)**

8. A force of 60 N gives an object an acceleration of 5 m/s². What is its mass? Show your work. **(Paragraph 6)**

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Answer Key

Vocabulary. force A push or a pull acting on an object, measured in newtons. **inertia** The tendency of an object to resist a change in its motion. **mass** The amount of matter in an object, measured in kilograms. **acceleration** A change in an object's speed or direction. **unbalanced** Forces that do not cancel out, so the object's motion changes. **newton** The unit of force, written N.

Eight Questions. Accept answers in the student's own words that carry the sense below. The place each answer comes from is noted, so you can check it was found rather than guessed.

1. The first law. No unbalanced force was acting on it, so it carried on doing what it was already doing. That is inertia. *(Paragraphs 1 and 2)*
2. Dev's board has more mass, so the same force produces less acceleration. F = m × a means acceleration goes down as mass goes up. *(Paragraph 3)*
3. His foot pushed the ground backward and the ground pushed him forward with an equal force in the opposite direction. Forces come in pairs. That is the third law. *(Paragraph 4)*
4. It pushes its own exhaust gas out one way, and the reaction to that push drives the rocket the other way. It does not need anything outside itself, because the force pair is between the rocket and its exhaust. *(Paragraph 5)*
5. The tendency of an object to resist a change in its motion. The story calls it stubbornness: a rolling board keeps rolling. *(Paragraph 2)*
6. Looking for F, so multiply. F = m × a = 4 × 3 = 12 N. The unit is required for the second mark. *(Paragraph 6)*
7. Looking for a, so divide. a = F ÷ m = 50 ÷ 10 = 5 m/s². The unit is required for the second mark. *(Paragraph 6)*
8. Looking for m, so divide. m = F ÷ a = 60 ÷ 5 = 12 kg. The unit is required for the second mark. *(Paragraph 6)*

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